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School of Mathematical Sciences

Computational Modelling of Multilateral Treaties

Stage 2

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Contents

List of Figures

List of Tables

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Principal Supervisor — Prof. Michael Bode Prof. Bode is Deputy Director of Capability Development at SAEF (Securing Antarctica's Environmental Future). His work centres on applied mathematics and conservation, with a focus on decision-making under uncertainty and mathematical biology. His previous work on Antarctic Treaty System (ATS) decision-making and conservation makes him an expert on consensus-based systems.

Associate Supervisor — Prof. Jonathan Rhodes Prof. Rhodes is Director of the Centre for Environment and Society. He specialises in spatial ecology, environmental governance, and the modelling of coupled human–natural systems, bridging quantitative models with real-world policy. His expertise gives him deep insight into the policy implications of environmental decisions, providing a grounding for the treaty-system applications explored in this research.

Chapter 1

Background and Literature Review

1.1 Introduction

Many international insitutions use consensus or unanimity as the voting system choice. This can implicitly caps what this institutions can actually achive. A consensus-based system earns broad legitimacy by granting equal veto power to every member. However, this gain in legitimacy trades off with slow adaptation, ultimately risking institutional inertia and decline. (Gray 2018). The Antarctic Treaty System (ATS) is one of those consensus-driven regimes, negotiated between a set of parties called consultative parties Secretariat of the Antarctic Treaty (2023). This group of countries, together with a second group lacking decision-making power (non-consultative parties), meets yearly at the Antarctic Treaty Consultative Meeting (ATCM). Before each meeting, culsultative parties may submit Working Papers (WP) and Information Papers (IP) up to 45 days before the ATCM starts Secretariat of the Antarctic Treaty (2023). During ATCMs the parties discuss Working Papers and produce outcomes only by consensus. These outcomes take the form of Measures, Decisions, or Resolutions Antarctic Treaty Consultative Meeting (1995). As international relations strain and Antarctic environmental pressures intensify, the risk of losing the capacity to act increases. In 2022, for the first time, the ATCM parties failed to reach consensus on the final report; even as activity has increased, the regime has become less responsive to challenges Gardiner et al. (2025).

In a system based on consensus is or utmost importance to uderstand why it succeds or fails, And which parties drive it or block it, respectively. Existing literature has described the ATS by rates and diversity of outputs, and stucturally by describing who co-aouthors with whom Larissa Lubiana Botelho et al. (2026) and Kumar et al. (2026). But there is no study where the a reconstruction of a directional party intention was reconstructed from the textual records on each issue over time. Even further, that type of measure has not been used to model the strategic behaviour that produces or breaks consensus. This thesis objective is to measure that and to build that model and asks if they contain the strategic intentions made by the consultative parties. It then asks whether these can model historical consensus dynamics.

The approach is a computational pipeline. First, it uses embedding techniques to discover topics, and NLI (Natural Language Inference) techniques for stance extraction. These would allow the construction of the preference measure. Second, it applies NLI to extract the opposition or aversion. It then testes the approach by comparing inputs (i.e. working papers) against observes outcomes. Finally, a Multi-Agent reinforcement learning is trained based on the previous outcomes to simulate the negotiation itself.

1.2 The Antarctic Treaty System as a consensus regime

In the ATS, Consultative Parties (CPs) reach outcomes exclusively by consensus. These discussions take place at the annual ATCM, where outcomes can take any of the following forms, per Secretariat of the Antarctic Treaty (2023, Rule 24):

- **Measures** are adopted texts containing provisions intended to become legally binding; these require approval by the relevant parties' legislative bodies.
- **Decisions** are adopted texts concerning internal organisational matters of the ATCM.
- **Resolutions** are non-binding recommendations or expressions of intent.

There are four types of input:

- **Working Papers** (Secretariat of the Antarctic Treaty (2023, Rule 48)): submitted by CPs and Observers, these require discussion at the ATCM and are the main source of outcomes. They are the only input from CPs that can produce an outcome; as such, they are the most likely place for each country to state its intentions.
- **Secretariat Papers** (Secretariat of the Antarctic Treaty (2023, Rule 49)): prepared and presented by the Secretariat, these produce organisational decisions as output, e.g. the budget.
- **Information Papers** (Secretariat of the Antarctic Treaty (2023, Rule 50)): supporting information for Working Papers or other discussions. They may be presented by CPs, Observers, Non-Consultative Parties (NCPs), and experts.

- **Background Papers** (Secretariat of the Antarctic Treaty (2023, Rule 51)): submitted by ATCM participants but not introduced during the meeting; they exist solely for the record.

Of all four, Working Papers are the only input uniquely submitted by CPs, requiring discussion and not confined to organisational matters. They are therefore the most likely place for countries to express their strategic intentions within this closed decision-making environment.

The consensus system makes the ATS a notable example of international cooperation; however, the same mechanism has become a major challenge as the number of CPs grows. Gardiner et al. (2025) show that the decision-making responsiveness of the ATS is declining; the unprecedented 2022 consensus failure is a clear example.

A notable case of contested consensus occurred in the late 1970s and 1980s, during the negotiations over mineral resources in Antarctica. The Convention on the Regulation of Antarctic Mineral Resource Activities (CRAMRA) was produced and then abandoned in favour of the Madrid Protocol (1991), which prohibited all mineral resource extraction in Antarctica (Jackson 2021). This well-documented episode provides a valuable canary test: a known case with documented positions against which any quantitative stance measure can be validated.

1.3 Quantitative and computational study of International institutions and treaties

In general the quantitative study of international institutions exist long before it was applied to Antarctica. Substantial literature maps the design of treaties and organisations into structured variables for example: their decision rules, dispute-settlement provisions, and enforcement mechanisms. Furthermore, these studies track these variable to to institutional performance and survival (Gray 2018).A more recent trend treats the *text* of international agreements and debates as data. They recover state positions from voting records and speeches (Baturo et al. 2017; Slapin and Proksch 2008). Both approaches show that preferences and design features of treaty regimes can be measured, not mereley described.

The ATS has been analysed quantitatively in two ways: Gardiner et al. (2025) does performance accounts that measure the regime in aggregate, counting inputs and outputs, and the time legally binding measures take to enter into force. It concludes that responsiveness to challenges in the Antarctic region is declining. The second methodology is structural (Larissa Lubiana Botelho et al. (2026) and Kumar et al.

(2026)): mapping who collaborates with whom through co-authorship networks of ATCM documents. Both analyses are very valuable, and in this project both results are going to be used as an external check: the network structure as a replication target for the authorship layer, and the outcome series as a prediction target in Chapter 2. Neither, however, extracts topics from WPs in an unsupervised manner, nor yields a directional, country- and topic-level, time-resolved measure of policy preference.

1.4 Measuring policy preference from political text

Positional measures extracted from political texts are a well-developed field with two paradigms. The first paradigm is scaling and ideal-point methods, which recover a latent positional dimension without a pre-specified axis, i.e. there is no need for a regulate vs permissive comparison. These methods tipically rely on word counts (i.e. frequencies) as their main underlying principle. This is the case of "Wordfish", introduced by Slapin and Proksch (2008) as an unsupervised positional scaling of party text. Lauderdale and Herzog (2016) addressed the problem that legislators talk about different things in different debates; they made a per-topic, per-debate scaling. On the other hand, Vafa et al. (2020) produced a probabilistic framework to place ideal points and the language of different actors. The closest dataset to the ATCM corpus was the one used by Baturo et al. (2017) to derive a state-level preference using United Nations (UN) debate speeches. Finally, a recent review by Parschan and Jakob (2025) condenses all these approaches in one paper. The second paradigm uses embeddings as its main tool. Rheault and Cochrane (2019) show that word-embedding geometry recovers ideological placement using parliamentary data. On the other hand, Rodriguez and Spirling (2022) provide a comprehensive analysis of what works and what does not when analysing political text using embedding techniques. They find embeddings a useful technique for those purposes but caution that the choice of embeddings strongly affects applied inference, and that the results must be validated. Finally, Licht (2023) and Chen et al. (2024) extend this approach to cross-lingual and multilingual settings.

Here, we propose to use both paradigms in the same pipeline but at different stages. This thesis differs from the ideal-point methods in an important way: ideal-point models recover a single stable latent position per actor, whereas the presented project requires a topic-conditioned stance on a given axis. The proposal in this thesis allows the parties to hold different positions on different topics through time. The topic structure is discovered using the embedding-based approach of Grootendorst (2022), which couples transformer representations with density clustering and class-based TF-IDF. The stance measure is then estimated with reference to the models of Burnham (2023) and Abercrombie and Batista-Navarro (2022). This thesis proposes to use discriminative transformer-NLI models, which take a **premise** and a **hypothesis**. These NLI models output one of the prescribed classes and $P(\text{entailment} \mid \text{hypothesis})$, then the class with the highest probability is selected. The final results can be validated directly using the CRAMRA canary test (Jackson 2021), or indirectly by checking which countries cluster around $(\text{topic}, \text{stance})$ pairs and comparing this to the Larissa Lubiana Botelho et al. (2026) network results.

1.5 Modelling consensus, coalition formation and multi-actor decision-making

There two traditional group of models how groups reach concensus. The first is the mosts ellegant but are the synchronization and opinion-dynamics models, like Kuramoto and opinion-changing-rate proposed by Leonardo L. Botelho (2022) which represent agreement as an coupled-oscillator convergence. These methods treat concensus as *convergence*. DeGroot (1974) average agents opinion iteratively using $x(t+1) = Wx(t)$ until they agree. There is also a bounded-version of this model debeloped by Hegselmann and Krause (2002). Another relevant model worth mentioning within this group is the stubborn agents (Friedkin and Johnsen 1990) where each agent does not fully update to match their neighbours, they stay, partially achored to their initial opinion. The aforementioned group of models does not contain strategic preference, no veto and no learning, therefore, it cannot simulate the bargaining behaviour that occurs in concensus based systems.

The second group of models uses social choice and game theory to model conseusus as a unanimity rule. Tsebelis (2002) develop a veto-players framework where every party has veto power, therefore, the status quo is only replace by new measures when $u_i(p) \geq u_i(q)$ for every Party i. Consequently, when the number of parties increase and the spread of their position grow, the set of adoptable policies decreases and the systems stability increases. This conclusion is the same reached by Gardiner et al. (2025) when analysing the ATS qualitatively.

1.6 Negotiation and coalition formation simulation using reinforcement learning

The strategic view of consensus developed above implies that, in consensus-based systems, the best way to replicate behaviour among many heterogeneous Parties, each holding a veto and bargaining over a status quo, is to use repeated strategic interaction.

In the specific case of the quantitative simulation of the ATS over the years, its size and complexity mean that the closed-form equilibria of analytic game theory are intractable. In these cases, multi-agent reinforcement learning (MARL) is the appropriate tool, precisely because it keeps the game-theoretic substance while relaxing this constraint. Formally, MARL is defined over stochastic (Markov) games, a generalisation of the repeated and normal-form games of Section 1.5. Similarly, its learning procedures approximate equilibrium behaviour from interaction rather than solving for it in closed form (Littman 1994). In this context, neural networks (NN) enter only as function approximators that allow this learning to scale to large state and action spaces.

Foundational MARL work covers multi-issue bargaining (Chang 2021), negotiating-team formation (Bachrach et al. 2020), bargaining under risk and asymmetric information (Schmid and Zotero CH3] 2020), and search-improved game-theoretic learning (Li et al. 2023). In this field, there are review papers by Gronauer and Diepold (2022) on MARL and by Sarkar et al. (2022) on coalition formation. Most recently, MARL agents based on Large Language Models (LLMs) have been used to model political negotiation directly: for example, Moghimifar et al. (2024) simulate coalition formation in parliament using text data from party manifestos. Bolleddu (2025) present a MARL system for consensus building and negotiation based on synthetic data. Kulkarni et al. (2025) use negotiated agreements with game-theoretic reasoning to predict dynamic coalition formation in Diplomacy (a well-known benchmark for this type of simulation). Finally, some other benchmarks for real negotiation data are starting to appear (Benac et al. 2026; Gu et al. 2025). However, none of the works mentioned above link an empirically extracted and validated country-by-topic preference structure to a strategic simulation of a real consensus treaty regime.

1.7 Summary of Literature Gap

The above literature review has a common arc at every step of the process: it goes from qualitative methods, to descriptive measures, to structural mapping and, finally, to learning-based methods. This proposal is confined to the quantitative elements of that arc, each of which leaves questions unanswered. Furthermore, although this work is intended to develop a framework that could be extended to other international institutions, it will focus on the ATS. In this scenario, the descriptive work shows what the regime outputs but not who produces what. Structural and party-network analyses are undirected and unattibuted. Text-based analysis recovers latent positions rather than topic-stance analysis through time. Finally, consensus has been simulated either as convergence or, where learning-based methods were used, on synthetic data or data other than the institution's own deliberative records. No prior work extracts a validated, signed, directional preference for each party over time from a consensus regime's own textual records. Not has any study used the extracted signed preferences per party to drive a strategic simulation in which agents have veto power and that is validated against realised outcomes. This research project intends to supply that coupling.

Chapter 2

Program and Design of the Research Investigation

2.1 Research Questions

The thesis addresses the gap mentioned above by answering these reserch questions:

Research Questions

RQ1. Can Parties' strategic positions be reconstructed from written records into a directional measure across topics and time?

RQ2. Does the reconstructed positions recover known history and predict outcomes?

RQ3. Can a multi-agent model parameterised by it reproduce consensus and test counterfactual decision rules?

2.2 Objectives, Methodology and Research Plan

Figure 2.1 shows the overall flow of the research project, the output of each stage, and how the objectives relate to one another. Throughout the following sections each chapter is described in detail.

2.2.1 Objective 1 / Chapter 1: The Support Layer

The **Aim** is to produce a defensible, directional measure of Parties' preferences over relevant topics through time, using documents the Parties submit themselves. For

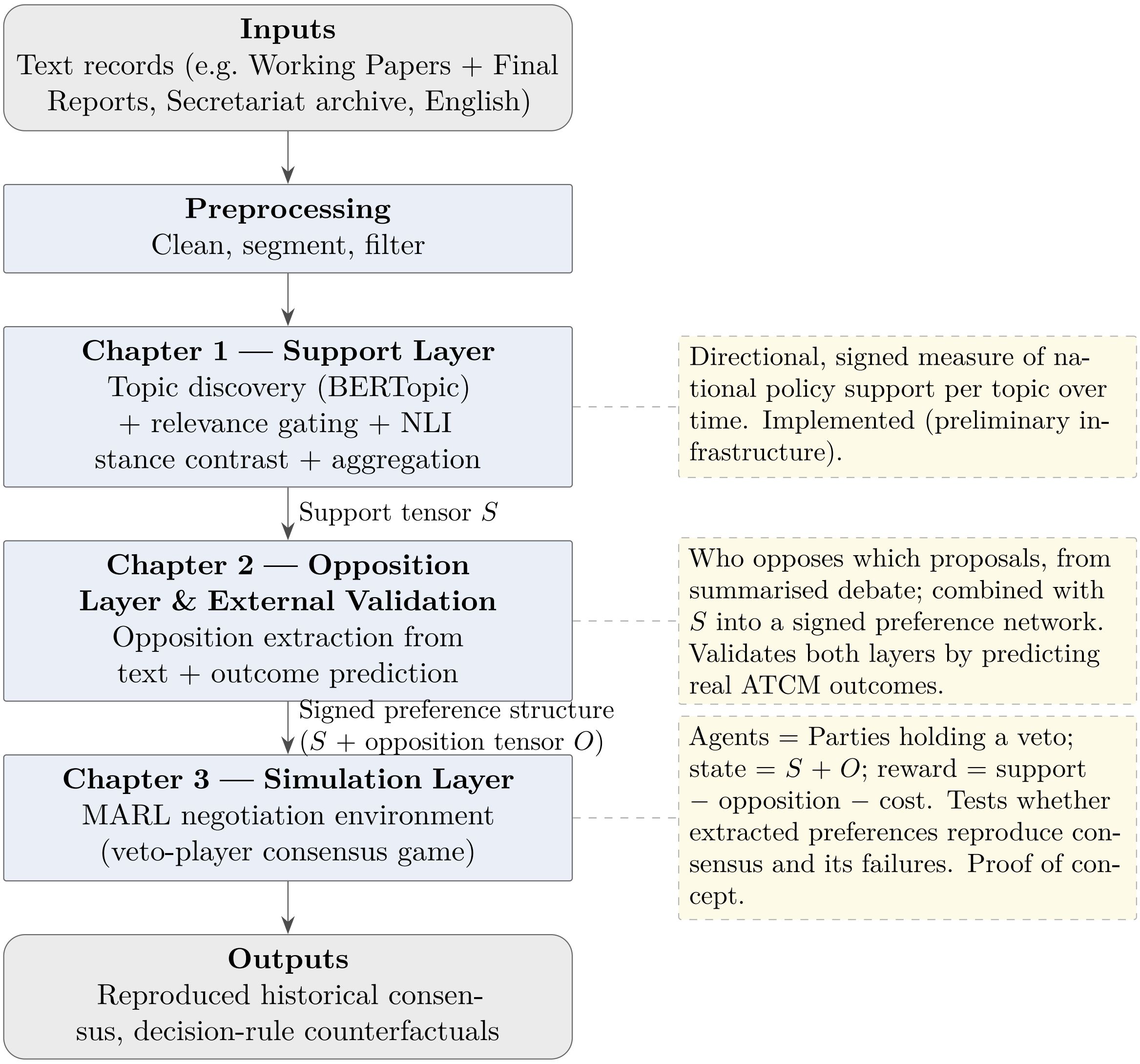

Figure 2.1: Thesis workflow: from text records to a validated consensus simulation. The support layer (Chapter 1) produces a signed position tensor S; the opposition layer (Chapter 2) adds an opposition tensor O and validates both against ATCM outcomes; the simulation layer (Chapter 3) tests whether the extracted preferences reproduce consensus and its failures.

the ATCM case, Working Papers (WPs) are the input data because, of all submitted documents, these are the only ones submitted by Parties or Observers (non-Secretariat) that require discussion and action. This makes WPs the most likely text to contain Parties' strategic intentions. The analysis uses English-language papers, since these represent the largest share of all WPs. It is also relevant that most of the available pre-trained embedding and NLI models were trained on English-language data.

The **approach** is a pipeline with multiple stages. The first set of stages applies BERTopic (Grootendorst 2022) to the whole corpus, producing a set of topics K with corresponding cluster centroids $\{\boldsymbol{\mu}_k\}_{k\in K}$, where each $\boldsymbol{\mu}_k \in \mathbb{R}^m$ in the m-dimensional embedding space. The second set of stages selects, for each document and topic, the passages relevant to that topic. Each document d is split into passages $P_d = \{p_1, \ldots, p_{n_d}\}$, and each passage p is embedded as $\mathbf{e}_p \in \mathbb{R}^m$. Relevance acts as a gating mechanism that filters out irrelevant passages: for each document d and topic k, every passage $p \in P_d$ is scored by its cosine similarity to the topic centroid as follows,

$$\rho_{p,k} = \cos(\mathbf{e}_p, \boldsymbol{\mu}_k) = \frac{\mathbf{e}_p \cdot \boldsymbol{\mu}_k}{\|\mathbf{e}_p\| \, \|\boldsymbol{\mu}_k\|}.$$

$P_{d,k}$ is the set of top scoring passages above a threshold. Then, the relevant sentences out of each $P_{d,k}$ are extracted using and Large Laguage Model (LLM). This sentences are concatenaited together and are scored as a set by the classifier. It produces one five-class distribution over ordinal classes that range from strong opposition to strong support. Formally, it is as follows

$$\pi_{d,k} = \mathrm{NLI}\Big(\mathrm{Extract}(P_{d,k})\Big) \in \Delta_5, \qquad \Delta_5 = \Big\{ \pi \in \mathbb{R}^5 \;\Big|\; \pi_\ell \geq 0,\ \textstyle\sum_{\ell=1}^{5} \pi_\ell = 1 \Big\}, \quad (2.1)$$

The signed stance of document d on topic k is the expected value of this distribution under an equal-spacing convention. This is:

$$s_{d,k} = \sum_{\ell=1}^{5} v_\ell \, \pi_{d,k}(\ell), \qquad v = \Big(-1,\, -\tfrac{1}{2},\, 0,\, \tfrac{1}{2},\, 1 \Big). \quad (2.2)$$

Since $\pi_{d,k}$ is a probability distribution and every $v_\ell \in [-1, 1]$, the stance $s_{d,k} \in [-1, 1]$ is a convex combination of the class values and therefore well defined on topic k's axis.

Finally, the previous result is aggregated to produce a *country* × *topic* × *year* stances over the Working Papers weighted by co-sponsorship credit:

$$S_{c,k,y} = \sum_{\substack{d \in D(c,y) \\ k \in K(d)}} w_{d,c} \, s_{d,k}, \quad (2.3)$$

where $D(c, y)$ is the set of Working Papers co-authored by country c in year y, $K(d)$ are the set of topics assigned to document d, and $w_{d,c}$ the attribution weight of that country in one document. Two attribution conventions are computed:

- full credit, $w_{d,c} = 1$, which means every co-sponsor receives the full position and which answers "who is on this side?".
- fractional credit, $w_{d,c} = 1/|A_d|$ with A_d the set of co-sponsors of d, which answers "who is driving this?".

For a fixed year, the slice $S_{\cdot,\cdot,y} \in \mathbb{R}^{C \times |K|}$ is a signed *country* $\times$ *topic* matrix, with C countries and $|K|$ topics. See Appendix C for full notation.

To measure directional stance on diplomatic text (2.1) represent a challenge give the hedging and similar prose in disting positioning. In other words, permissive and restrictive positions on a regulatory question share same safeguard vocabulary, therefore, scoring a single hypothesis collapses toward neutrality. Threfore, stance is recovered as a *contrast* between entail and contradiction positions. Specifically, for each topic k, structured hypotheses anchor a for-pole h_k^+, a neutral one h_k^0, and an against one h_k^- on the topic's contestable axis. These $(h_k, premise)$ pairs are input into NLI classifier (Laurer et al. 2024). The resulting scores for *entailment* and *contradiction* are subtracted. The classifier's results are cross-checked against a generative model used as a secondary instrument (Le Mens and Gallego 2024).

Validation strategy The construct will be validated using three independent checks, each targeting a different stage of the pipeline. A topic validation check compares the discovered topics against the Secretariat's categories and the WP in each category. A historical canary uses the minerals/CRAMRA negotiations of the late 1970s and 1980s as documented ground truth. The testing asserts if the pipeline recovers the known directional opposition in those papers. Finally, a network comparison check reconstructs the co-authorship structure reported by Larissa Lubiana Botelho et al. (2026) from the authorship layer.

Figure (2.2) below shows a summarize overview of the steps described above.

2.2.2 Objective 2 / Chapter 2: The Opposition Layer and External Validation

The **aim** of this part of the research project is to extract a directional opposition signal. This opposition signal has two sides. The first is opposition to the issue itself; it is not the inverse of the support-layer result, but rather the *veto signal* against a given

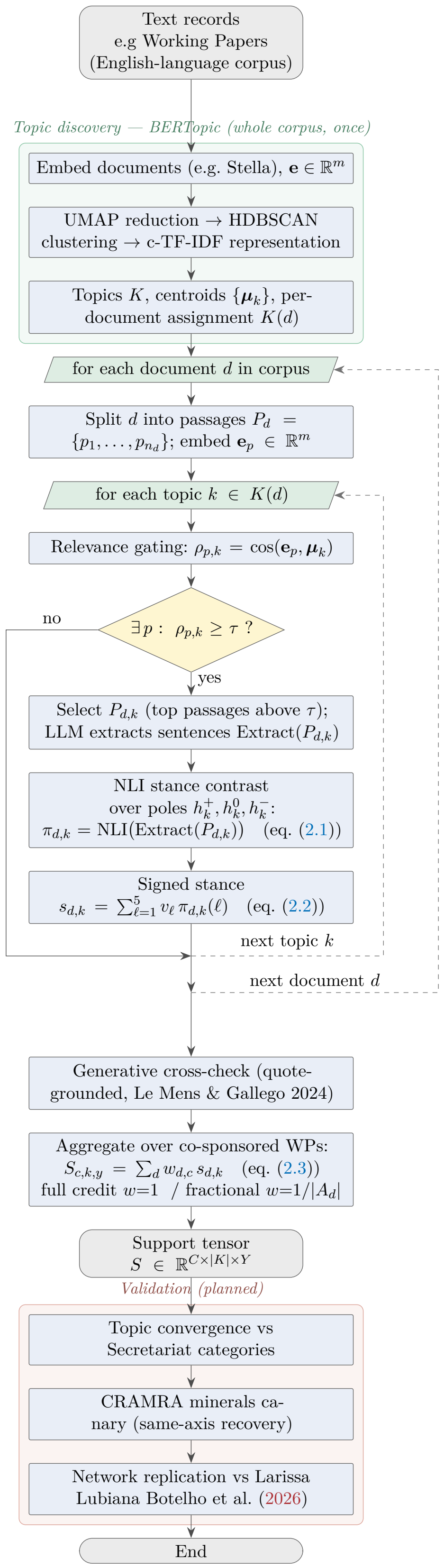

Figure 2.2: Chapter 1 support-layer pipeline: from text records through BERTopic topic discovery, relevance gating, NLI stance contrast, and aggregation to the country×topic×year support tensor S.

proposal. The second side is secundary exploratory result: a measure of a country's general propensity to reject proposals within a given ATCM. As with the support layer, this chapter builds a pipeline and, for the specific case of the ATS, the ATCMs' final reports are used as text input. Given the scarcity of text in which countries veto a proposal, the attribution is expected to be harder than for the support layer. Each extracted statement yields a *(country, target, polarity)* triple, with rationale and source quote. Taking the topics K from Chapter 1 and mapping targets back to them yields an opposition tensor $\mathbf{O}$ with the same Country × Topic × Year structure. At this point, there is a pair $(\mathbf{S}, \mathbf{O})$ of tensors that constitutes a signed preference network. These two tensors are the inputs to the third chapter.

External validation. The combined structure (the tensor pair $(\mathbf{S}, \mathbf{O})$) is tested for predictive content against observed outcomes (see 1.2) and the compilation by Gardiner et al. (2025), for the case of the ATS. This gives Chapter 2 an independent research question, test the results so far, and produces a quantitative description. The novelty in this work is the directional signed signal derived from the institution's own input records and outcomes.

The main risks in this chapter are the sparsity of text that can be directly attributed to an input, and the identification of the country that vetoes any proposal. A further limitation is the influence of external geopolitics, as in the case of the consensus breakdown in 2022 due to the war in Ukraine (Gardiner et al. 2025). The cause of this type of exogenous influence is unlikely to be reflected in the text data of an institution not directly related to such issues.

Figure 2.3 sketches the opposition layer pipeline and its external-validation step for the ATCM case study.

Objective 3 / Chapter 3: The Simulation Layer

Aim. Simulate the ATCM consensus process with country-agents whose state is parameterised by the empirically derived support and opposition layers, reproduce historical trajectories, and examine institutional counterfactuals. Because the consensus rule makes every Party a veto player (Tsebelis 2002), the natural representation is a strategic bargaining game over a status quo rather than an averaging dynamic; this is the substantive reason a multi-agent reinforcement-learning formulation is adopted in place of the synchronisation models of Section **??**.

Environment and agents. We specify a custom negotiation environment in which proposals are presented one at a time — each proposal being a single Working Paper

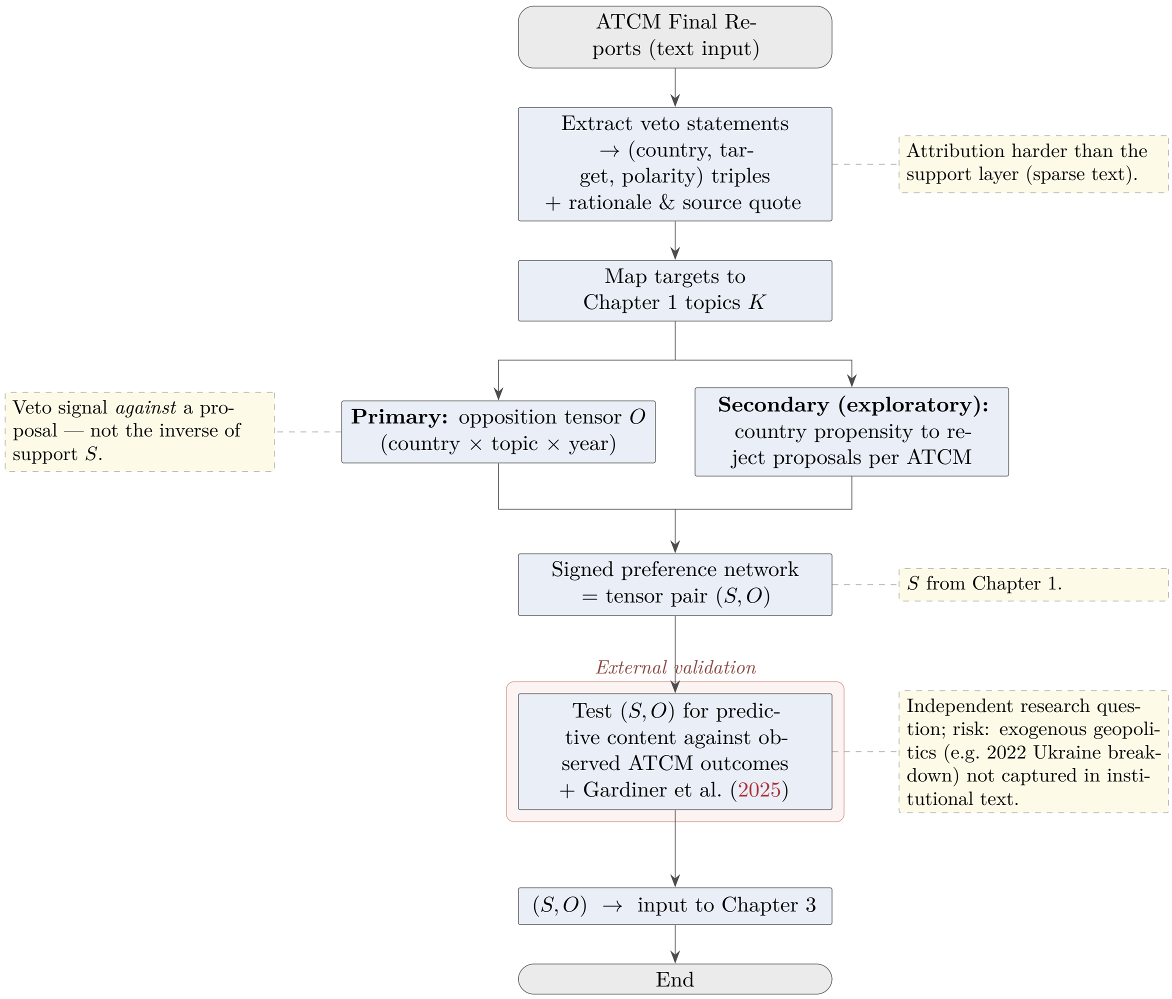

Figure 2.3: Chapter 2 opposition layer sketch: from text data to opposition tensor O, forming the signed preference network (S, O) together with the Chapter 1 support tensor S. It is validated against observed ATCM outcomes before feeding Chapter 3.

d on topic k in year y, drawn from the historical record — and decided by consensus, with one learning agent per Consultative Party (Gronauer and Diepold 2022; Schmid and Zotero CH3] 2020). Each agent's state fuses its support vector (Chapter 1), its opposition history (Chapter 2), and slow-moving country covariates such as consultative status and accession year, used as context features rather than as an independent contribution. An agent's action on a presented proposal is binary: to object or not, $o_i \in \{0, 1\}$. Under the consensus rule the proposal is adopted only if no Party objects,

$$\nVdash_{\text{adopt}} = \prod_j \left(1 - o_j\right), \tag{2.4}$$

so that a single objection vetoes adoption. The reward to agent i is then

$$R_i = \alpha\, S_{i,k,y} \nVdash_{\text{adopt}} - \beta\, O_{i,k,y} \nVdash_{\text{adopt}} - \gamma\, o_i, \tag{2.5}$$

where the first term rewards adoption of outcomes the Party supports, the second penalises adoption of outcomes it opposes, the third charges a diplomatic cost for casting an objection, and $\alpha, \beta, \gamma \geq 0$ trade the three off; reward is measured relative to the status quo, normalised to zero. Because adoption (2.4) is the product of all Parties' non-objection, each agent holds a genuine veto, and consensus emerges from the strategic trade-off between blocking an opposed outcome and bearing the cost of objection — not from imposed convergence.

Validation and counterfactuals. The model is validated by out-of-sample prediction of consensus or failure for held-out years and by an ablation that removes the opposition layer to quantify its contribution. Counterfactual analysis (RQ3) then examines alternative decision rules — for example qualified majority in place of unanimity, which relaxes (2.4) — against the effectiveness metrics of Gardiner et al. (2025). This chapter grounds its agents in empirically extracted, validated positions from the institution's own record, distinguishing it from MARL consensus in simulated negotiation environments (Bolleddu 2025), manifesto-grounded language-model agents (Moghimifar et al. 2024), and abstract synchronisation (Larissa Lubiana Botelho et al. 2026). It is the highest-risk chapter and is framed as a proof of concept, contingent on Chapters 1 and 2 passing validation.

2.3 Objective 3 / Chapter 3: The Simulation Layer

2.4 Paper 3 — Integrated Simulation Framework

2.5 Data Sources and Ethics

The primary data sources for this project are:

- Antarctic Treaty Consultative Meeting (ATCM) final reports and working papers (publicly available at https://www.ats.aq/).
- State-level foreign policy position datasets (e.g., UN voting records).
- Published network and voting data from the international relations literature.

All data are publicly available institutional records; no human-subjects research is involved and no ethics clearance is required beyond standard QUT research integrity protocols.

2.6 Timeline

Table 2.1: Indicative project timeline

Phase	Activity	Target Date
1	Literature review & theoretical framework	Month 6
2	Paper 1: data collection & analysis	Month 12
3	Paper 1: writing & submission	Month 15
4	Paper 2: model development & calibration	Month 18
5	Paper 2: writing & submission	Month 21
6	Paper 3: simulation framework development	Month 27
7	Paper 3: writing & submission	Month 30
8	Thesis integration & write-up	Month 36
9	Thesis submission	Month 38

2.7 Expected Contributions

1. A formal computational model of consensus dynamics in the ATS, the first of its kind at this level of institutional specificity.

2. A transferable agent-based simulation framework applicable to any consensus-based multilateral treaty.

3. New theoretical constructs (affinity/aversion indices) operationalising previously qualitative political science concepts.

4. Policy-relevant insights into treaty design features that improve stability under strategic pressure.

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Appendix A

Supplementary Material

Stance classifier (Chapter 1). For topic k, let h_k^+, h_k^0, h_k^- be the for-, neutral-, and against-pole hypotheses. For a pooled passage set d and hypothesis h, the debiased evidence is the null-premise-corrected log-ratio

$$\lambda(d,h) = \mathrm{clip}\Big(\log P(\mathrm{entail} \mid d,h) - \log P(\mathrm{entail} \mid \varnothing, h),\ \pm 5\Big),$$

which removes each hypothesis's marginal acceptability (a directed pointwise-mutual-information / DCPMI debiasing). The three-pole distribution combines entailment of each pole with contradiction of its opposite,

$$\tilde{\pi}_{d,k} = \mathrm{softmax}\begin{pmatrix} \lambda_{\mathrm{ent}}(h_k^+) + \lambda_{\mathrm{con}}(h_k^-) \\ \lambda_{\mathrm{ent}}(h_k^0) \\ \lambda_{\mathrm{ent}}(h_k^-) + \lambda_{\mathrm{con}}(h_k^+) \end{pmatrix},$$

which is expanded to the five ordinal classes $\pi_{d,k}$ and reduced to the scalar stance by the expected value of Equation (1).

Appendix C: Notation

Indices and sets.

c, i	countries (Consultative Parties), $c \in \{1, \ldots, C\}$
k	topic, $k \in K$; $\lvert K\rvert$ is the number of topics
y	year
d	document (Working Paper)
p	passage (chunk) of a document
ℓ	ordinal stance class, $\ell \in \{1, \ldots, 5\}$
$D(c, y)$	Working Papers co-sponsored by country c in year y
$K(d)$	topics assigned to document d
P_d	passages of document d, $P_d = \{p_1, \ldots, p_{n_d}\}$
$P_{d,k}$	passages of d selected as relevant to topic k
A_d	co-sponsors of document d

Embedding space.

m	embedding dimensionality (full Stella space)
$\mathbf{e}_p$	embedding of passage p, $\mathbf{e}_p \in \mathbb{R}^m$
$\boldsymbol{\mu}_k$	centroid of topic k, $\boldsymbol{\mu}_k \in \mathbb{R}^m$
$\rho_{p,k}$	passage–topic relevance, $\rho_{p,k} = \cos(\mathbf{e}_p, \boldsymbol{\mu}_k)$
τ	relevance selection threshold

Stance and hypotheses.

h_k^+, h_k^0, h_k^-	for-, neutral-, against-pole hypotheses for topic k
$\lambda(d, h)$	null-premise-debiased entailment log-ratio (DCPMI)
$\tilde{\pi}_{d,k}$	three-pole distribution (Appendix C, stance classifier)
$\pi_{d,k}$	five-class ordinal distribution, $\pi_{d,k} \in \Delta_5$
Δ_5	probability simplex, $\{\pi \in \mathbb{R}^5 : \pi_\ell \geq 0,\ \sum_\ell \pi_\ell = 1\}$
v	class values, $v = (-1, -\frac{1}{2}, 0, \frac{1}{2}, 1)$
$s_{d,k}$	signed stance of d on topic k, $s_{d,k} \in [-1, 1]$

Aggregation and tensors.

$w_{d,c}$	attribution weight: 1 (full credit) or $1/\lvert A_d\rvert$ (fractional)
$S_{c,k,y}$	support of country c on topic k in year y, Eq. (2.3)
$S_{\cdot,\cdot,y}$	year-y stance slice, $S_{\cdot,\cdot,y} \in \mathbb{R}^{C\times\lvert K\rvert}$